LE NGUYEN DUY LUAN

Mobile Developer | On-device AI / Computer Vision
Nha Trang, Khanh Hoa, Vietnam | 0774 653 337 | luanle1702@gmail.com
Portfolio | GitHub | LinkedIn

Professional Summary

Detail-oriented Mobile Developer specializing in Native Android (Kotlin, Jetpack Compose) and On-device AI/Computer Vision (TensorFlow Lite). Experienced in building performance-optimized, offline-first mobile applications with real-time ML capabilities and cloud-backed sync (Firebase, Supabase). Strong problem-solving skills and dedication to writing clean, maintainable code. Proven ability to bridge AI research and mobile deployment — from model training to TFLite on-device inference.

Education

Nha Trang University 2022 – 2026
Bachelor of Information Technology *Nha Trang, Vietnam*

- **GPA**: 3.25/4.0
- **Graduation Project**: Lung X-ray Disease Classification using Segmentation-guided Masking — Score: 9.1/10

Technical Skills

Languages: Java, Kotlin, JavaScript, Python, SQL
Mobile: Android SDK, Jetpack Compose, CameraX, Room DB, WorkManager, Coroutines & Flow
AI / Computer Vision: TensorFlow Lite, MediaPipe, MLKit (Pose Estimation), YOLO, OpenCV, Gemini API, Groq API (LLaMA)
Backend Services: Firebase (Auth, Firestore, Storage), Supabase, REST API design, OkHttp
Tools: Git/GitHub, Android Studio, Gradle, ADB (Android Debug Bridge), KSP, VS Code, Figma (basic)
AI-Assisted Dev: Google Antigravity, OpenAI Codex
Languages: Vietnamese (Native) – English (Intermediate - proficient in technical documentation)

Projects

VocabLensAI — Native Android English Learning Platform | *Personal Project*

- **Tech Stack**: Kotlin, Jetpack Compose, Coroutines/Flow, Room DB, CameraX, TensorFlow Lite, Gemini API, Firebase, Supabase.
- Developed an offline-first Android application from scratch using Kotlin, Jetpack Compose, MVVM, and clean code practices.
- Solved the problem of passive vocabulary memorization by implementing real-time on-device object detection (CameraX + TensorFlow Lite), integrating Gemini AI to generate contextual learning stories from scanned vocabulary.
- Addressed the "forgetting curve" problem in language learning by designing a custom SM-2 Spaced Repetition engine that automatically schedules reviews at optimal intervals, backed by a hybrid Room DB/Firestore sync system to ensure reliable offline data sync.
- Built a real-time multiplayer PvP Arena using Firebase Firestore with an ELO rating calculator, alongside a chess-style game review UI for match accuracy analysis and AI feedback.

Lung X-ray Trustworthy Classifier (Android + Research) | *Graduation Project, NTU* 9.1/10

- **Tech Stack**: Python, TFLite, U-Net (ResNet50), YOLOv12m-cls, Kotlin, Jetpack Compose, Groq API.
- Translated academic research into a production-grade native Android app: exported a 2-stage AI pipeline (U-Net segmentation → YOLOv12m-cls classification) to TFLite for fully offline, on-device inference across 4 pathology classes (COVID-19, Viral Pneumonia, Lung Opacity, Normal).
- Engineered a segmentation-guided median-fill masking layer to eliminate attention leakage and shortcut learning outside lung regions; validated with EBPG (+12% vs. raw baseline) and IoU localization across 300 annotated COVID-19 cases.
- Built an OOD safety gate that auto-rejects non-X-ray inputs via lung-area ratio thresholding (valid range: 8–72%), preventing the model from generating false-confident results on out-of-distribution images — a critical safety design for medical AI prototypes.

- Integrated dual-mode XAI (Occlusion Sensitivity on-device + Grad-CAM++ via PyTorch server) and a context-aware LLM assistant (Groq/LLaMA 3.1) grounded in Fleischner Society guidelines, with automatic offline fallback for reliable demo environments.
- Implemented native PDF research report export (Android PdfDocument API — zero third-party libs), benchmarked 5 backbone architectures (DenseNet121, EfficientNet-B3, ResNet50, ViT-B/16, YOLOv12m-cls) with full trustworthiness metrics displayed in-app.

AI Fitness Coach (Android) | *Personal Project*

- **Tech Stack**: Kotlin, Jetpack Compose, CameraX, MediaPipe/MLKit (Pose Estimation), Room DB, Text-To-Speech (TTS).
- Developed an offline-first Android app (MVVM) providing a comprehensive ecosystem with an 800+ exercise library and an AI-driven personalized workout recommendation engine.
- Addressed the injury risk of unsupervised home workouts by implementing camera-based pose estimation (MediaPipe) to extract 33 joint coordinates and calculate 3D vector angles in real-time (60 FPS).
- Built a custom rule-based feedback engine with Text-To-Speech integration (audio alerts) and automated rep counting, caching progress 100% offline via Room DB.